

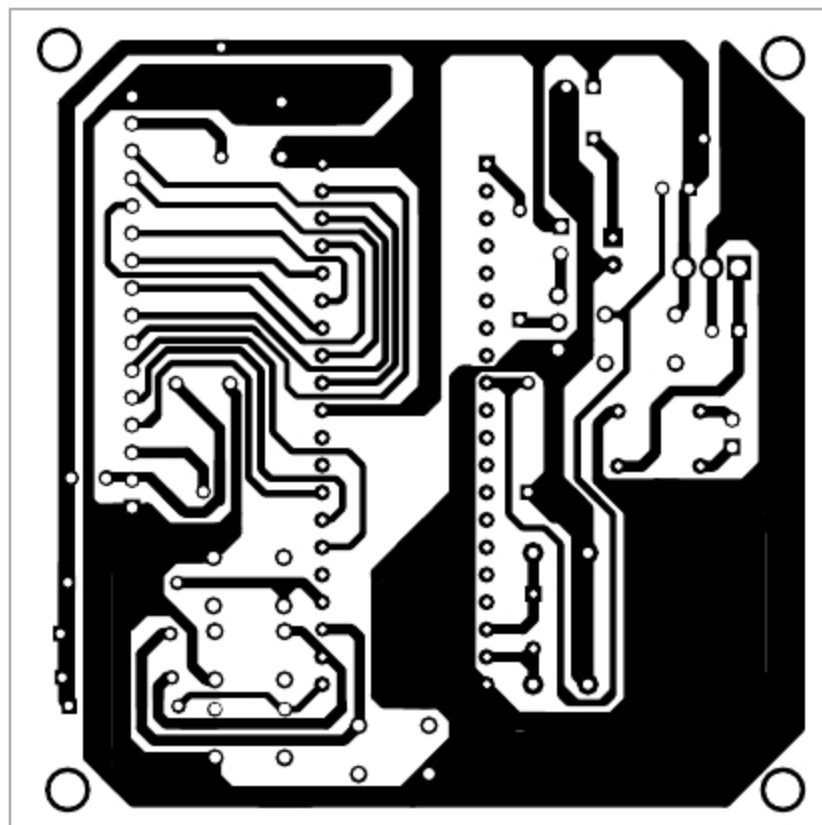


Fig. 2: Actual-size PCB for fastest finger first

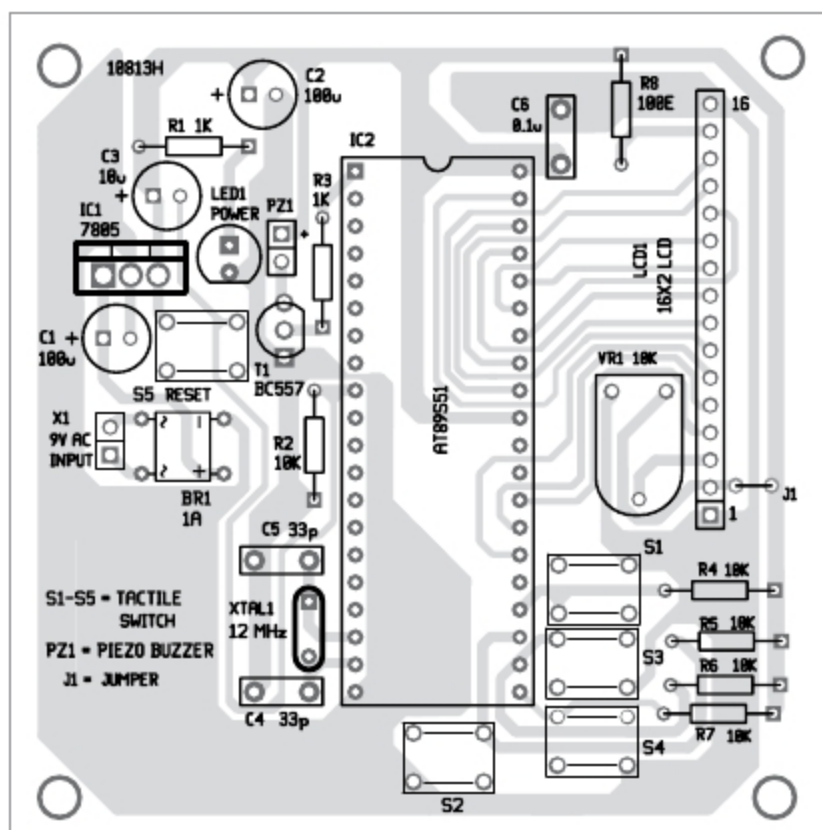


Fig. 3: Components layout for the PCB

Whenever a switch is pressed, logic 0 is given as input to the respective input pin. And, LCD1 displays the corresponding name of the user. Preset VR1 is used to control the contrast voltage for LCD1, while resistor R8 limits the current flow to LED back-light of LCD1.

An actual-size PCB layout for the MCU-based fastest finger first is shown in Fig. 2 and its components layout in Fig. 3. Assemble the project as per the circuit diagram. Burn the given hex file into the MCU. Power on the circuit, and the LCD will display “fastest finger first” message. If the text is not visible, adjust VR1 till the text is clearly visible. **EFY**

Whenever the circuit is switched on, or reset switch S5 is pressed, the device starts its operation with a beep sound and displays “fastest finger first” message in the first line of LCD1. Then, LCD1 shows “1 2 3 4” in the first line and waits until the user presses a switch.

Whenever a switch is pressed, the corresponding name is displayed in the second line of LCD1. The names appear in the sequence their switches are pressed. After the last (fourth) switch is pressed, “!Press! !Reset!” message is displayed on LCD1 followed by a beep sound. The cycle can be started again by pressing S5.

Software

The program is written in embedded C language using Keil software. The code also works on other 8051 (89C51, 89C52, 89S51, etc) compatible MCUs. The compiled hex file can be burned into the MCU using any suitable programmer. At EFY Lab, USB 8051 programmer was used to burn the hex code.

Construction and testing

EFY Note

The source code of this project is available for free download at source.efymag.com



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-Henry Ford

